

Qatar-2b

R-1980

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160606 · created 2026-08-07T00:41:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457545.70377 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.028
Transit depth (Rp/Rs)^2	0.0003 +/- 0.0096 [%]
Orbital Inclination (inc)	87.23 +/- 1.69
Airmass coefficient 1 (a1)	0.962 +/- 0.035
Airmass coefficient 2 (a2)	0.033 +/- 0.016
Scatter in the residuals of the lightcurve fit is	3.7 %
Best Comparison Star	#3 - [370, 476]
Optimal Method	PSF photometry
Transit Duration (day)	0.0529 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.002 ± 0.028 vs 0.16327 (deviation 5.07σ)
Duration fit vs published	0.0529 d vs 0.0767 d (deviation 6.53σ)
Mid-time O-C	0.3 min
Depth SNR	0.1
Residual scatter	3.7 %

Light curve

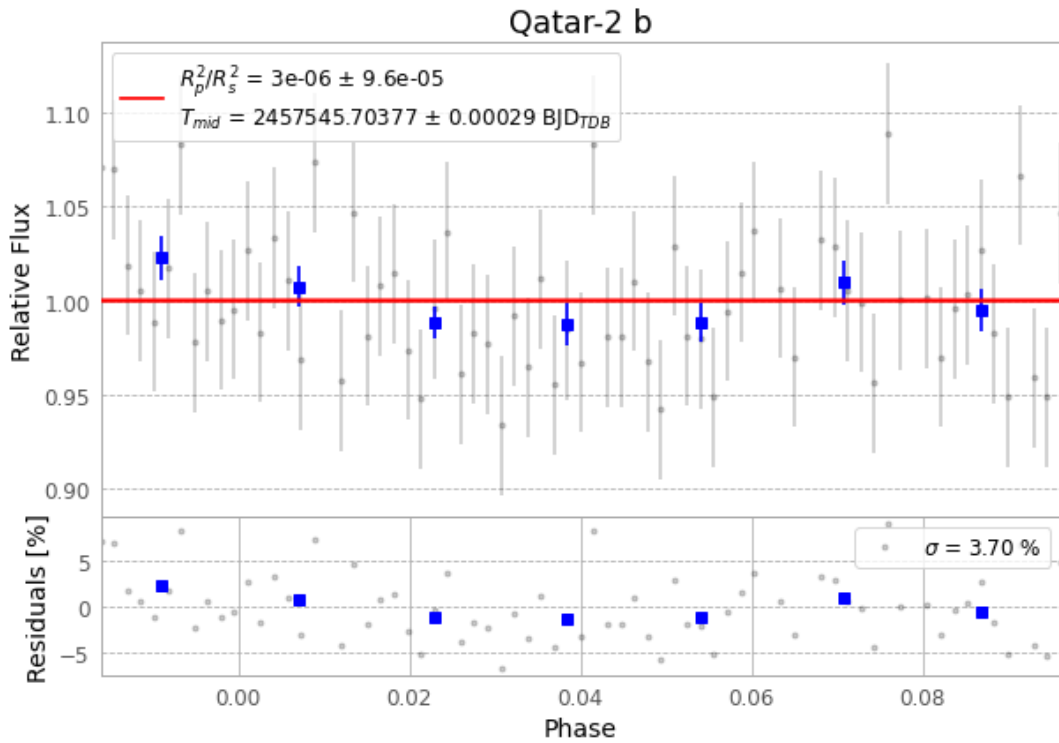


Figure 1 — FinalLightCurve_Qatar-2 b_2016-06-05.png (SHA-256 2b3249285310bb6a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [277, 445], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 25c64da61272a01b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

73 FITS frames (48 MB), Qatar-2160606041512.FITS → Qatar-2160606075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160606.log (eed2841f2ca5...), FinalLightCurve_Qatar-2 b_2016-06-05.png (2b3249285310...), FinalLightCurve_Qatar-2 b_2016-06-05.csv (aaee4c7ca34c...)

Compute resources

Wall time 106.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:41Z.